DS323: Machine Learning
Project Report
2nd Semester, 2025-2026

Predicting Academic Performance Levels of University Students Using Eating Behavior Patterns

By:

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1. Problem Statement

University students often develop irregular and unhealthy eating habits due to academic pressure and lifestyle changes. These behaviors, such as skipping meals, consuming fast food, and consuming high amounts of caffeine, may negatively affect cognitive performance and academic outcomes.

However, the relationship between eating habits and student performance is not sufficiently explored using data-driven methods.

2. Project Aim & Objectives

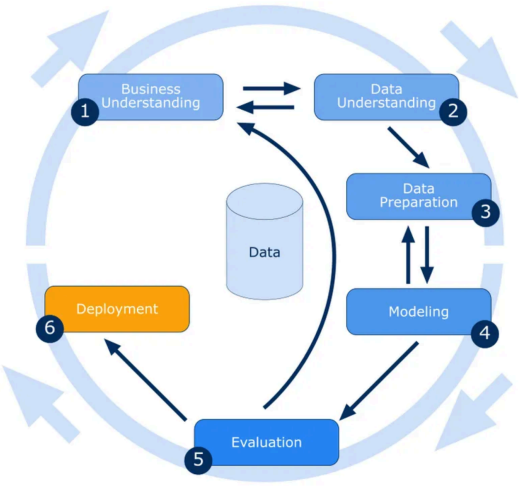
2.1. Project Aim

The project aim is to develop a machine learning classification model that predicts students' academic outcomes based on their eating behaviors, in order to understand how these habits influence academic performance.

2.2. Project Objectives

- Collect and analyze data related to students' eating behaviors and lifestyle habits.
- Preprocess the dataset by handling missing values and outliers.
- Perform EDA to identify patterns and relationships between eating behaviors and academic outcomes.
- Build and compare three machine learning classification models.
- Evaluate the performance of the models by using appropriate performance metrics.
- Identify the most significant factors that influence students' academic outcomes.

3. Methodology



CRISP-DM Methodology Flowchart.

The **CRISP-DM** (Cross Industry Standard Process for Data Mining) methodology was adopted in this study to examine the relationship between students' eating habits and their academic performance. This methodology is based on the standard CRISP-DM framework [1] and provides a structured approach consisting of six sequential phases.

3.1 Business Understanding

The objective of this study is to analyze the impact of students' eating behaviors (e.g., meal patterns, fast-food consumption, and caffeine intake) on academic performance, and to develop a classification model for prediction.

3.2 Data Understanding

The dataset was explored to understand its structure, identify missing values, and examine key variables related to eating habits and academic outcomes.

3.3 Data Preparation

The data was cleaned and transformed by handling missing values and outliers, encoding categorical variables, and selecting relevant features for modeling.

3.4 Modeling

Several classification models were developed using training and testing datasets to identify patterns and predict academic performance.

3.5 Evaluation

The models were evaluated using performance metrics such as accuracy, precision, recall, and F1-score to determine the best-performing model.

3.6 Deployment

The final results were presented to provide insights into how eating habits influence students' academic performance

4. AI Ethics

During the data preprocessing and modelling stages, missing values in features will be identified and handled to ensure data quality, while irrelevant identifiers will be removed as they have no predictive value. The dataset will also be examined for outliers; however, since the data mainly consists of ordinal categorical values (0, 1, 2, 3), no significant outliers are expected, which supports stable model performance. These steps help ensure that the model remains fair, accurate, and trustworthy when predicting student academic performance based on eating habits.

5. Dataset Description

The dataset used in this project was collected through an online survey distributed to university students across different universities in Saudi Arabia. The purpose of this dataset is to investigate the relationship between students' eating behaviors and their academic performance. The dataset consists of 200 instances, each representing a student, and 15 attributes, including both input features and the target variable.

The dataset features describe students' eating habits, lifestyle, and academic context. These features include:

Academic_Year: Student's academic year.
Working_Status: Indicates whether the student is working (No, Part-time, Full-time).
Meals_per_day: Number of meals consumed daily.
Breakfast: Frequency of eating breakfast.
Fastfood_consumption: Frequency of fast food consumption.
Studying_food: Type of food consumed while studying.
Caffeine_consumption: Number of caffeinated drinks consumed daily.
LateNight_eating: Frequency of eating late at night.
FruitsandVegs_consumption: Frequency of consuming fruits and vegetables.
Sleep_avg: Average number of sleep hours per night.
Exercise_avg: Frequency of exercise.
Study_avg: Average number of study hours per day.
Late_studying: Frequency of studying late at night.
GPA_range/n: High (5.0-4.5), Medium (4.5-3.5), and low (3.5 and below).

These variables were selected to capture aspects that may influence students' academic performance, with a particular focus on dietary patterns and daily habits. The target variable in this dataset is GPA_range, which represents students' academic performance categorized into three classes: High (4.5–5.0), Medium (3.5–4.49), and Low (Below 3.5). Therefore, the problem addressed in this project is a multi-class classification problem [2].

The dataset contains a combination of integer (int64) and floating-point (float64) data types. All categorical responses collected from the survey were manually encoded as numerical

values (e.g., 0, 1, 2, 3) to ensure compatibility with machine learning algorithms [3], without using external encoding techniques. Overall, the dataset is self-collected and reflects the real-world behaviour of university students, making it suitable for this analysis.

6. Exploratory Data Analysis

Before applying machine learning models, the dataset was explored and preprocessed to ensure data quality and consistency. Initially, the dataset was examined for missing values, where a small number of missing entries were identified in features such as Meals_per_day, Breakfast, Fastfood_consumption, Caffeine_consumption, and LateNight_eating. These missing values were removed during preprocessing to prevent any negative impact on model performance. Additionally, the ID column was dropped because it only serves as row numbering and does not contribute to the prediction task.

Following this, the dataset was cleaned and prepared to ensure consistency across all feature values. Outlier detection was considered; however, since the dataset consists mainly of ordinal categorical variables within a small numerical range, no significant outliers were observed. Therefore, no outlier removal methods were required. Similarly, normalization or scaling was not applied, as the data does not contain large numerical ranges or continuous variables that require rescaling.

To better understand the data, exploratory analysis was performed on the target variable. The results indicated that the dataset is relatively balanced across the three classes, as shown in Figure B.1, which is beneficial for training machine learning models, as it reduces bias toward any specific class.

A correlation heatmap was generated to analyze the relationships between the features and the target variable; the results indicate generally weak-to-moderate correlations, suggesting that multiple factors collectively influence students' academic performance rather than a single dominant feature, as shown in Figure C.1.

Finally, the first 10 rows of the preprocessed dataset were displayed to verify that the data is clean, consistent, and ready for model training, as shown in Figure C.2.

7. Model Selection

Since the target variable represents students' academic outcome categories, this study is formulated as a supervised multiclass classification problem. Based on the characteristics of the dataset and the project objective, three models were selected: Logistic Regression, Support Vector Machine (SVM), and Random Forest.

1-Logistic Regression was chosen as a baseline model due to its simplicity, efficiency, and interpretability. It is appropriate for classification tasks and provides a clear understanding of how the input variables may relate to academic outcomes.

2-Support Vector Machine (SVM) was selected because it performs effectively on small to medium-sized datasets and is well-suited for handling complex decision boundaries. This makes it a strong choice for the current dataset, in which students' eating and lifestyle behaviours may have nonlinear effects on academic performance.

3-Random Forest was included because of its ability to capture non-linear relationships and variable interactions. It is also robust to noise and useful for identifying the most influential features within the dataset.

Together, these models offer a meaningful comparison between a linear interpretable approach, a powerful margin-based classifier, and a flexible ensemble method. Therefore, they are appropriate for predicting students' academic outcomes based on eating behaviours and related lifestyle factors.

8. Model Estimation

Three classification models were implemented and trained: Logistic Regression, SVM, and Random Forest. The dataset was split into 70% for training and 30% for testing to ensure a reliable evaluation of the models on unseen data. GridSearchCV was utilized with 5-fold cross-validation to tune the hyperparameters and identify the optimal settings for each classifier. To further enhance the Random Forest model, the StandardScaler technique was applied to its features and the model was retrained to analyze the impact of feature scaling on its predictive capabilities.

9. Model Evaluation

The performance of all three models was evaluated using multiple metrics, including accuracy, precision, recall, and F1-score, as these metrics together provide a more comprehensive and unbiased assessment of classification model performance than accuracy alone [4]. A detailed summary of all metrics for each model is presented in Table D.1, while the overall test accuracy and model status are summarized in Table D.2.

Logistic Regression achieved the highest test accuracy of 52.5% and was identified as the best-performing model. The classification report shows that class 0 (Low GPA) had the best results with a precision of 0.62, recall of 0.68, and F1-score of 0.65. Class 1 (Medium GPA) had the weakest performance with a precision of 0.40, recall of 0.32, and F1-score of 0.35, suggesting the model struggled most to distinguish medium-performing students. The confusion matrix for Logistic Regression is shown in Figure D.1. Since the training accuracy of 63.2% is below 70% and the test accuracy is also low, the model is considered to suffer from underfitting, meaning it is too simple to fully capture the underlying patterns in the data.

Support Vector Machine (SVM) achieved a test accuracy of 49.2% with a training accuracy of 93.4%. The significant gap between training and test accuracy indicates overfitting, where the model memorized the training data but failed to generalize to unseen instances. The low precision and recall values across all three classes further confirm that SVM's predictions were unreliable despite its high training performance.

Random Forest achieved a test accuracy of 45.8% with a perfect training accuracy of 100%, which is the clearest sign of overfitting among all three models, and it was therefore identified as the worst model. The StandardScaler was applied to its features and the model was retrained; however, the test accuracy remained unchanged at 45.8%, as shown in Figure D.2, indicating that scaling did not improve the model's ability to generalize on this dataset.

Overall, the evaluation results across all metrics confirm that Logistic Regression is the best model for this task, despite its modest accuracy. The consistently weak performance across all three models suggests that eating behavior patterns alone have limited predictive power for academic performance, which highlights the need for richer and larger datasets in future work.

10. Deployment

The Students' Outcomes Based on Eating Behaviors application was deployed using Streamlit Community Cloud. The project files were uploaded to GitHub, including the Streamlit application file, the dataset, and the requirements file. The project output is shown in Appendix E.

The deployed application allows users to explore the dataset, view visualizations, compare machine learning models, and predict students' academic performance levels based on their eating behavior and lifestyle patterns.

The main Streamlit file is mlproject.py, which contains the application interface and prediction logic. The dataset file is used to display and analyze the data inside the application, while requirements.txt includes the Python libraries required to run the app on Streamlit Cloud.

The application interface includes input widgets such as sliders, select boxes, and buttons. These widgets allow users to enter values for features such as meals per day, breakfast frequency, fast food consumption, caffeine consumption, late-night eating, fruits and vegetables consumption, sleep average, exercise average, study average, and late studying.

After entering the required values, the user clicks the prediction button, and the trained machine learning model predicts the student's academic performance level as Low, Medium, or High.

11. Conclusion

This project aimed to predict the academic performance of university students based on their eating behaviors and daily lifestyle habits using machine learning classification models. Three models were developed and evaluated: Logistic Regression, Support Vector Machine (SVM), and Random Forest. Among the three, Logistic Regression achieved the highest test accuracy of 52.5%, while Random Forest was identified as the worst-performing model with a test accuracy of 45.7%. Grid Search with Cross-Validation was then applied to all models to find the optimal hyperparameter values. After tuning, Random Forest showed improvement, reaching a test accuracy of 49.1%, while Logistic Regression and SVM did not benefit from tuning on this dataset. These results suggest that eating behavior patterns alone provide limited predictive power for academic performance, which aligns with findings in the literature indicating that academic outcomes are influenced by a combination of behavioral, psychological, and environmental factors [5].

This project benefits the community by raising awareness of the connection between students' daily habits and their academic outcomes. Universities and academic advisors can use insights from this work to design targeted wellness programs that encourage healthier eating, improved sleep, and regular exercise among students. Additionally, this work contributes a self-collected dataset reflecting real-world behaviors of university students in Saudi Arabia, which can serve as a foundation for future research.

For future work, the dataset size should be expanded beyond 200 instances to improve model generalizability and reduce the risk of underfitting. Additional features such as mental health indicators, social support, and financial stress could be incorporated to capture a more complete picture of what influences academic performance. Furthermore, more advanced models such as ensemble stacking or neural networks could be explored to potentially achieve higher predictive accuracy.

References

[1] C. Schröer, F. Kruse, and J. M. Gómez, "A systematic literature review on applying CRISP-DM process model," *Procedia Computer Science*, vol. 181, pp. 526–534, 2021. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1877050921002416

[2] T. M. Mitchell, *Machine Learning*. New York, NY, USA: McGraw-Hill, 1997. [Online]. Available: http://www.cs.cmu.edu/~tom/mlbook.html

[3] F. Pedregosa *et al.*, "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011. [Online]. Available: https://jmlr.org/papers/v12/pedregosa11a.html

[4] M. Berková, L. Konderla, and J. Závada, "Accuracy, precision, recall, F1-score, or MCC? Empirical evidence from advanced statistics, ML, and XAI for evaluating business predictive models," *Journal of Big Data*, vol. 12, no. 1, p. 72, 2025. [Online]. Available: https://link.springer.com/article/10.1186/s40537-025-01313-4

[5] Z. Alalawi, A. Al-Razgan, F. Al-Twairesh, and H. Alshaya, "Contextualizing the current state of research on the use of machine learning for student performance prediction: A systematic literature review," *Engineering Reports*, vol. 5, no. 12, e12699, 2023. [Online]. Available: https://onlinelibrary.wiley.com/doi/full/10.1002/eng2.12699

Appendices

Appendix A: Survey Questions

The following questions were used to collect the dataset:

1. What is your academic year?
2. Are you currently working while studying?
3. How many meals do you usually eat per day?
4. How often do you eat breakfast?
5. How often do you consume fast food?
6. What type of food do you usually eat while studying?
7. How many caffeinated drinks do you consume per day?
8. How often do you eat late at night?
9. How often do you eat fruits or vegetables?
10. How many hours do you sleep per night on average?
11. How often do you exercise?
12. How many hours do you study per day on average?
13. How often do you study late at night?
14. What is your current GPA range?

Appendix B: Dataset Sample

A sample of the dataset after preprocessing is shown below:

	Academic_Year	Working_Status	Meals_per_day	breakfast	fastfood_consumption	studying_food	caffine_consumption	latenight_eating	FruitsandVegs_consumption	sleep_avg	excersies_avg\n	study_avg\n	late_studying	GPA_range\n
0	3	1	1.0	1.0	1.0	0	3.0	3.0	3	0	2	1	3	2
1	2	1	0.0	3.0	1.0	4	2.0	3.0	1	1	0	3	3	2
2	2	0	1.0	2.0	1.0	0	2.0	0.0	1	2	2	0	0	1
3	2	0	1.0	1.0	1.0	1	1.0	1.0	2	2	0	2	3	1
4	2	0	0.0	3.0	2.0	4	0.0	1.0	0	2	0	1	1	2

Figure B.1 Dataset sample after preprocessing

Appendix C: Additional Visualization

Additional graphs generated during exploratory data analysis are presented below:

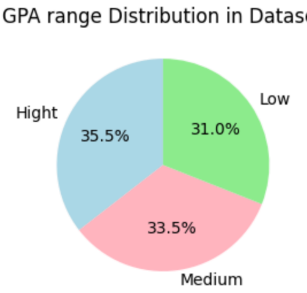


Figure C.1 Target column distribution

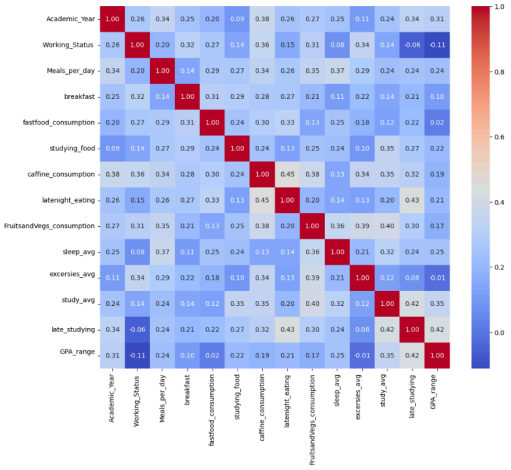


Figure C.2 Correlation heatmap

Appendix D: Model Building and Evaluation Results

ML Algorithms	Models' Performance			
	Accuracy	Precison	Recall	F1-Score
Logistic Regression	0.53	0.51	0.53	0.52
SVM	0.49	0.51	0.49	0.50
Random Forest	0.46	0.49	0.46	0.46

Table D.1 Models Performance Summary

Model	Training Accuracy	Test Accuracy	Model Status
Logistic Regression	0.63	0.53	Underfitting
SVM	0.83	0.49	Overfitting
Random Forest	1	0.46	Overfitting

D.2 Models Performance Metrics

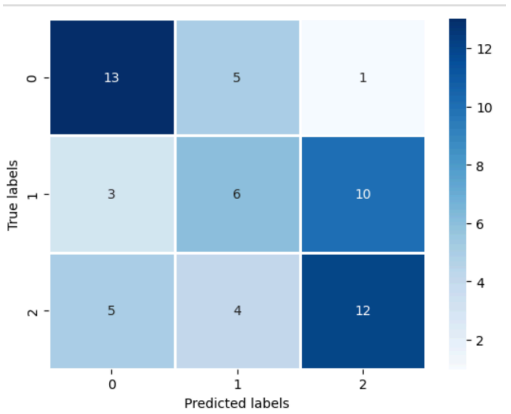


Figure D.1 Confusion Matrix for Logistic Regression (Best Model)

```
Training Accuracy before scaling: 1.0
Testing Accuracy before scaling : 0.4576271186440678

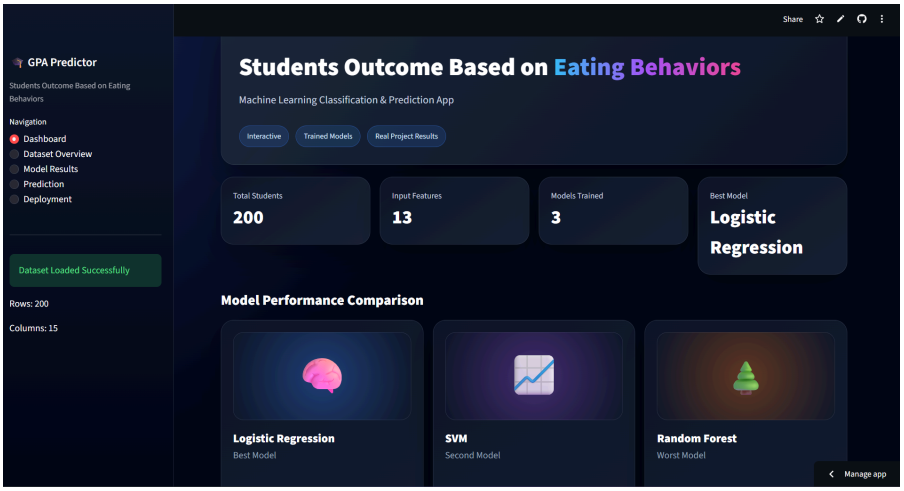
Training Accuracy after Pre-processing using StandardScaler :  1.0
Test Accuracy after Pre-processing using StandardScaler :  0.4576271186440678
```

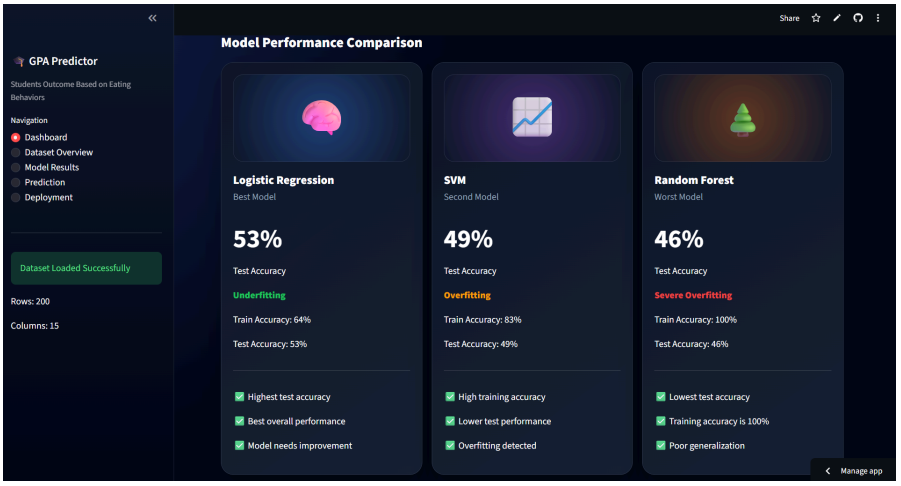
Figure D.2 Random Forest Model (Worst Model) before and after Scaling

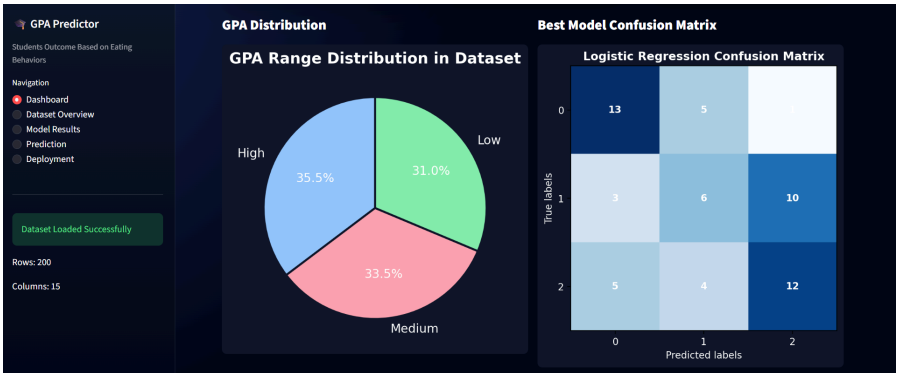
```
Logistic Regression - Default Test Accuracy :  0.5254237288135594
Logistic Regression - Tuned Test Accuracy   :  0.4745762711864407
SVM - Default Test Accuracy :  0.4915254237288136
SVM - Tuned Test Accuracy   :  0.4067796610169492
Random Forest - Default Test Accuracy :  0.4576271186440678
Random Forest - Tuned Test Accuracy   :  0.4915254237288136
```

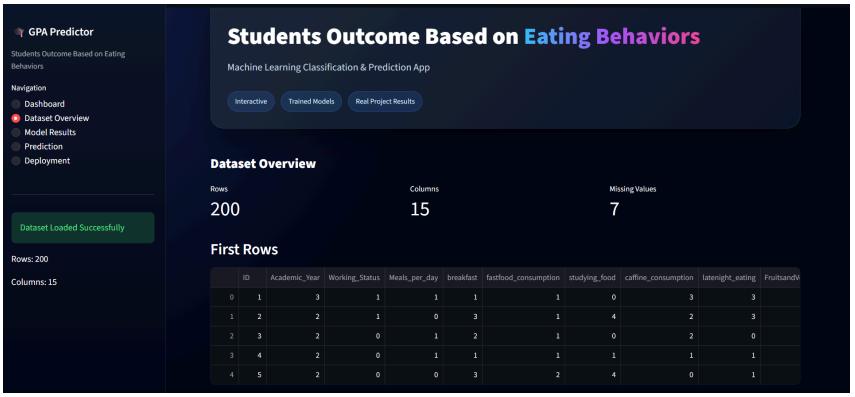
Figure D.3 Default and Tuned Test Accuracy Comparison for the three models

Appendix E:Deployment

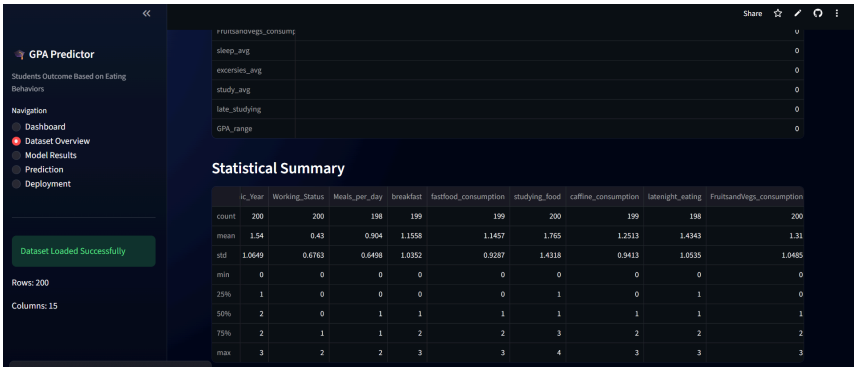

GPA Predictor
Students Outcome Based on Eating Behaviors
Navigation
Dashboard
Dataset Overview
Model Results
Prediction
Deployment
Dataset Loaded Successfully
Rows: 200
Columns: 15
Share
Students Outcome Based on Eating Behaviors
Machine Learning Classification & Prediction App
Interactive
Trained Models
Real Project Results
Total Students
200
Input Features
13
Models Trained
3
Best Model
Logistic Regression
Model Performance Comparison
Logistic Regression
Best Model
SVM
Second Model
Random Forest
Worst Model
Manage app

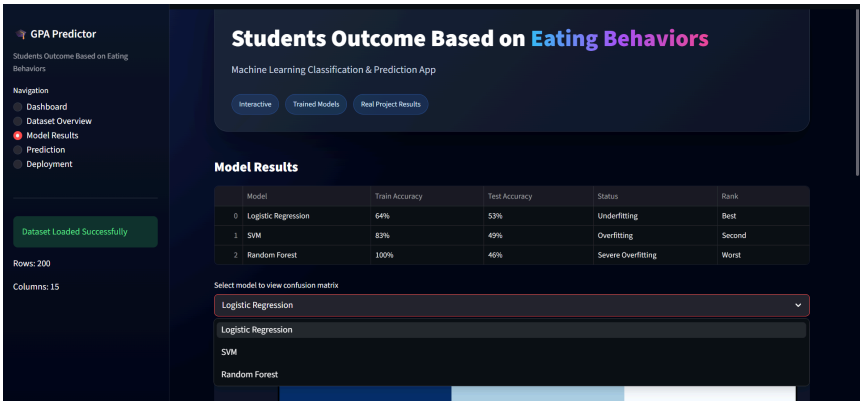

GPA Predictor
Students Outcome Based on Eating Behaviors
Navigation
Dashboard
Dataset Overview
Model Results
Prediction
Deployment
Dataset Loaded Successfully
Rows: 200
Columns: 15
Share
Model Performance Comparison
Logistic Regression
Best Model
53%
Test Accuracy
Underfitting
Train Accuracy: 64%
Test Accuracy: 53%
Highest test accuracy
Best overall performance
Model needs improvement
SVM
Second Model
49%
Test Accuracy
Overfitting
Train Accuracy: 83%
Test Accuracy: 49%
High training accuracy
Lower test performance
Overfitting detected
Random Forest
Worst Model
46%
Test Accuracy
Severe Overfitting
Train Accuracy: 100%
Test Accuracy: 46%
Lowest test accuracy
Training accuracy is 100%
Poor generalization
Manage app

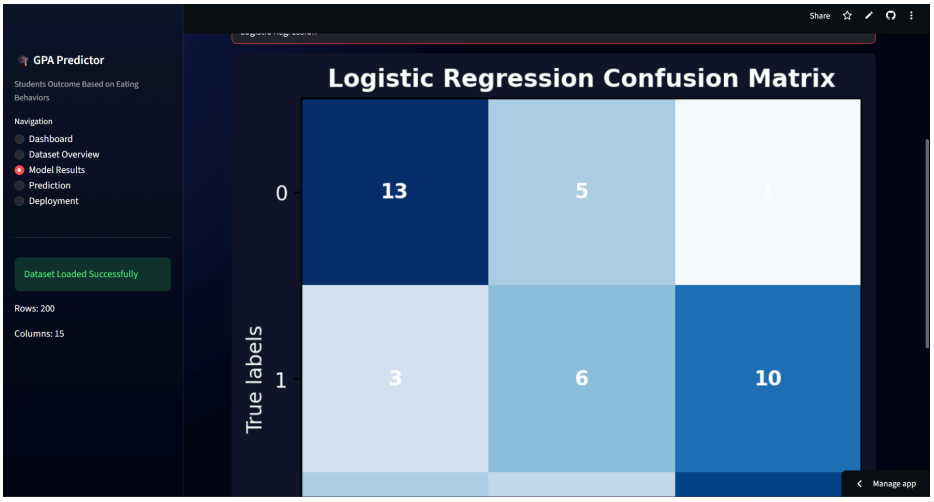

GPA Predictor
Students Outcome Based on Eating Behaviors
Navigation
Dashboard
Dataset Overview
Model Results
Prediction
Deployment
Dataset Loaded Successfully
Rows: 200
Columns: 15
GPA Distribution
GPA Range Distribution in Dataset
High
Low
Medium
35.5%
31.0%
33.5%
Best Model Confusion Matrix
Logistic Regression Confusion Matrix
True labels
Predicted labels
13
5
3
6
10
5
4
12

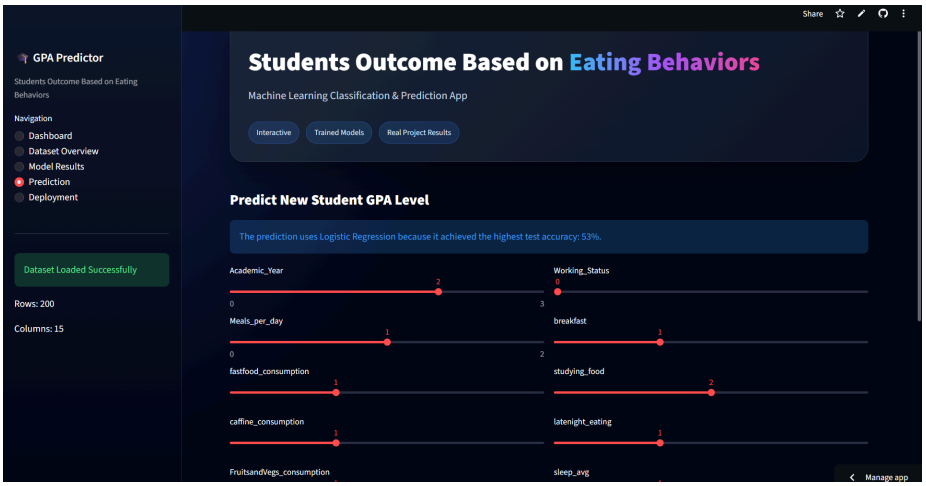

GPA Predictor
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Model Results
Prediction
Deployment
Dataset Loaded Successfully
Rows: 200
Columns: 15
Students Outcome Based on Eating Behaviors
Machine Learning Classification & Prediction App
Interactive
Trained Models
Real Project Results
Dataset Overview
Rows
200
Columns
15
Missing Values
7
First Rows

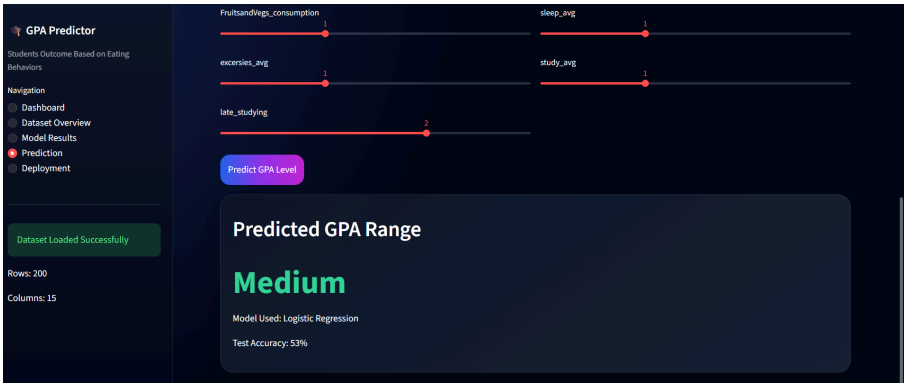

GPA Predictor
Students Outcome Based on Eating Behaviors
Navigation
Dashboard
Dataset Overview
Model Results
Prediction
Deployment
Dataset Loaded Successfully
Rows: 200
Missing Values
Missing Values
ID 0
Academic_Year 0
Working_Status 0
Meals_per_day 2
breakfast 1
fastfood_consumption 1
studying_food 0
caffine_consumption 1
latenight_eating 2


GPA Predictor
Students Outcome Based on Eating Behaviors
Navigation
Dashboard
Dataset Overview
Model Results
Prediction
Deployment
Dataset Loaded Successfully
Rows: 200
Columns: 15
Share
sleep_avg 0
excersies_avg 0
study_avg 0
late_studying 0
GPA_range 0
Statistical Summary


GPA Predictor
Students Outcome Based on Eating Behaviors
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Model Results
Prediction
Deployment
Dataset Loaded Successfully
Rows: 200
Columns: 15
Students Outcome Based on Eating Behaviors
Machine Learning Classification & Prediction App
Interactive
Trained Models
Real Project Results
Model Results
Model
Train Accuracy
Test Accuracy
Status
Rank
0 Logistic Regression 64% 53% Underfitting Best
1 SVM 83% 49% Overfitting Second
2 Random Forest 100% 46% Severe Overfitting Worst
Select model to view confusion matrix
Logistic Regression
Logistic Regression
SVM
Random Forest


GPA Predictor
Students Outcome Based on Eating Behaviors
Navigation
Dashboard
Dataset Overview
Model Results
Prediction
Deployment
Dataset Loaded Successfully
Rows: 200
Columns: 15
Logistic Regression Confusion Matrix
True labels
0
1
13
5
1
3
6
10
Share
Manage app


GPA Predictor
Students Outcome Based on Eating Behaviors
Navigation
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Dataset Overview
Model Results
Prediction
Deployment
Dataset Loaded Successfully
Rows: 200
Columns: 15
Students Outcome Based on Eating Behaviors
Machine Learning Classification & Prediction App
Interactive
Trained Models
Real Project Results
Predict New Student GPA Level
The prediction uses Logistic Regression because it achieved the highest test accuracy: 53%.
Academic_Year
Working_Status
Meals_per_day
breakfast
fastfood_consumption
studying_food
caffine_consumption
latenight_eating
FruitsandVegs_consumption
sleep_avg
Share
Manage app



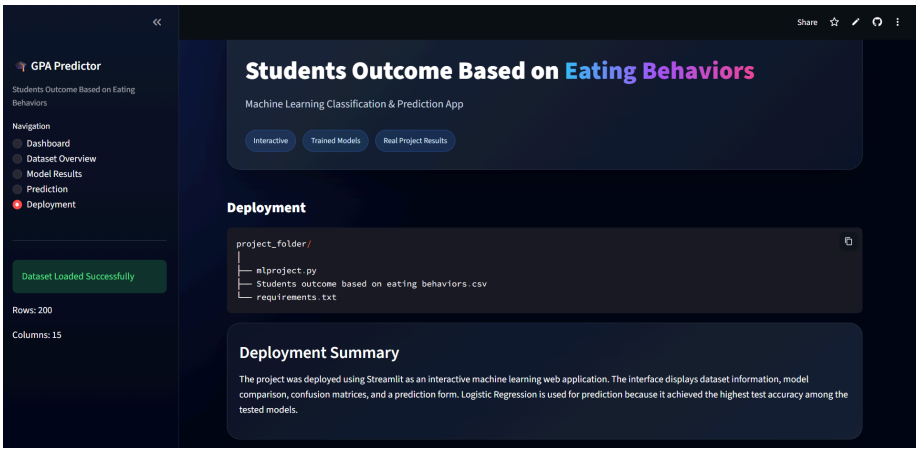


Figure E.1 Deployment Output

Appendix F: Links

Survey Link: Students outcome based on eating behaviors☕🍩. – Fill out form

Deployment Link:

https://mlproject-2wqyuvbdaamcmszln7ydq6.streamlit.app/